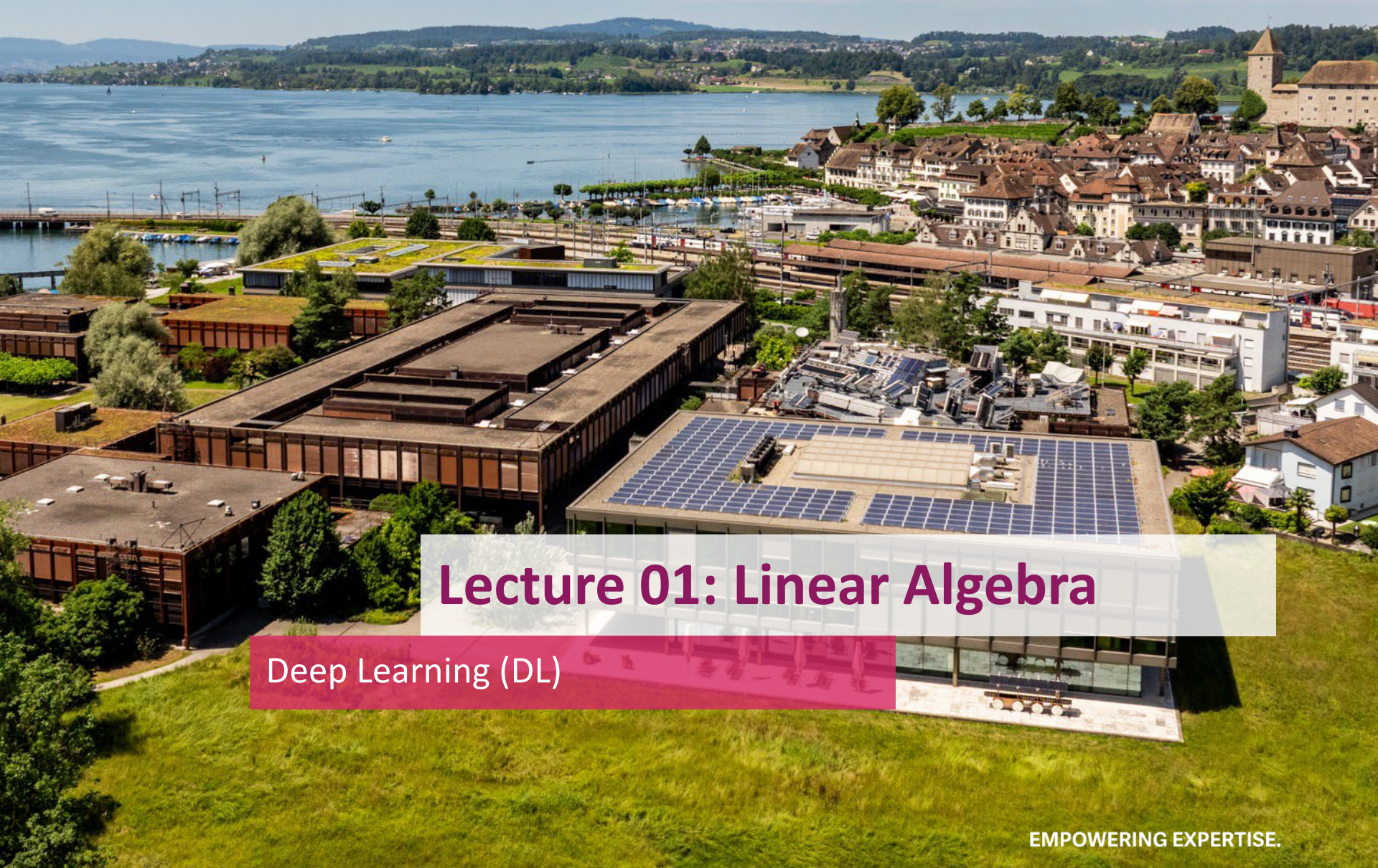




# Lecture 01: Linear Algebra

Deep Learning (DL)

EMPOWERING EXPERTISE.

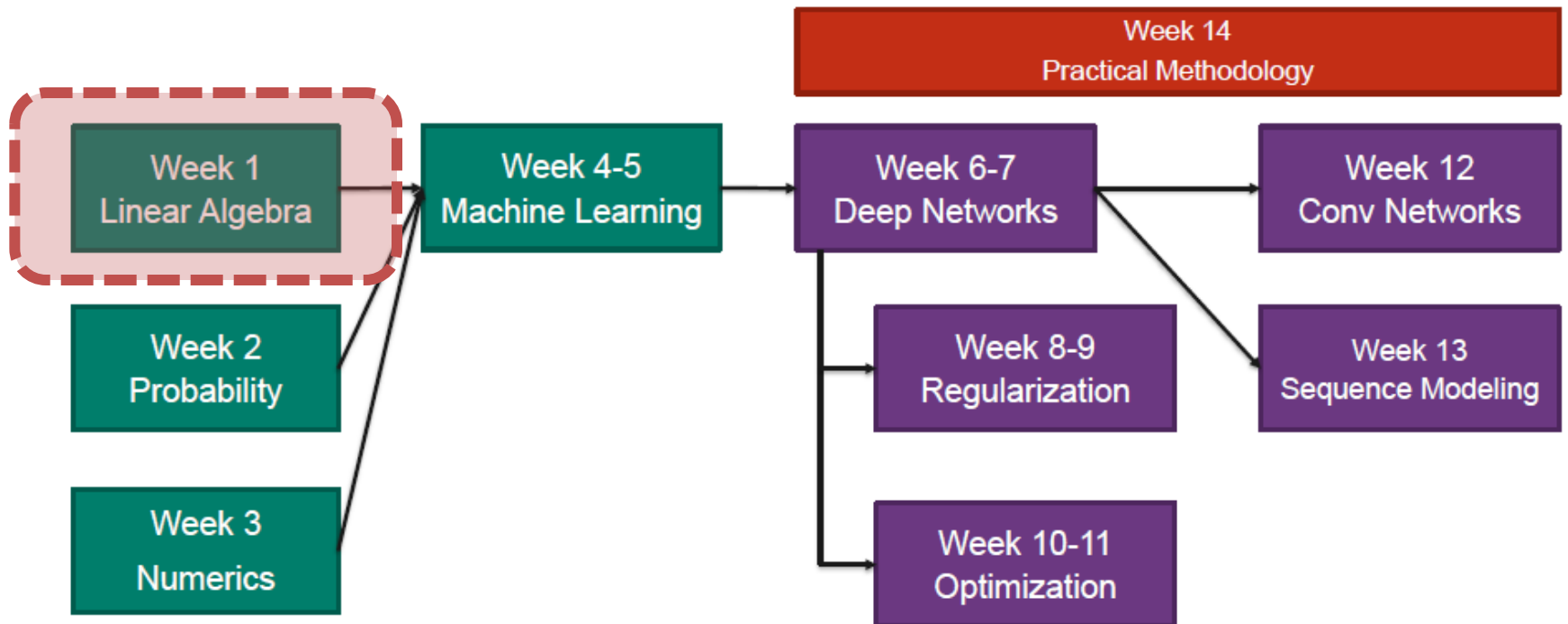


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16 February 2026



# Semester Plan



# Agenda

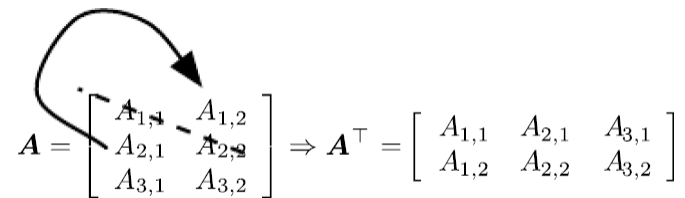
- Fundamentals of vectors and matrices
- Special matrices
- Matrix decompositions
- Moore-Penrose-Pseudoinverse
- The Trace and the Determinant



# Scalars, Vectors, Matrices and Tensors

- Scalars
  - A single number
- Vector
  - A **1D array** of numbers
  - Usually: column vectors
- Matrices
  - A **2D array** of numbers
  - One can think of a matrix as a collection of column vectors
  - One can think of a matrix as a collection of row vectors
  - Transposing a matrix is shown on the right
- Tensors
  - A 3D (or higher) array of numbers
  - The element of A at the coordinates (i,j,k) is denoted as  $A_{i,j,k}$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$



$$\mathbf{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \mathbf{A}^T = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

| Dimensions | Example                                                                                                                                         | Terminology |   |   |        |   |   |   |   |   |                                            |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---|---|--------|---|---|---|---|---|--------------------------------------------|
| 1          | <table border="1"><tr><td>0</td><td>1</td><td>2</td></tr></table>                                                                               | 0           | 1 | 2 | Vector |   |   |   |   |   |                                            |
| 0          | 1                                                                                                                                               | 2           |   |   |        |   |   |   |   |   |                                            |
| 2          | <table border="1"><tr><td>0</td><td>1</td><td>2</td></tr><tr><td>3</td><td>4</td><td>5</td></tr><tr><td>6</td><td>7</td><td>8</td></tr></table> | 0           | 1 | 2 | 3      | 4 | 5 | 6 | 7 | 8 | Matrix                                     |
| 0          | 1                                                                                                                                               | 2           |   |   |        |   |   |   |   |   |                                            |
| 3          | 4                                                                                                                                               | 5           |   |   |        |   |   |   |   |   |                                            |
| 6          | 7                                                                                                                                               | 8           |   |   |        |   |   |   |   |   |                                            |
| 3          | <table border="1"><tr><td>0</td><td>1</td><td>2</td></tr><tr><td>3</td><td>4</td><td>5</td></tr><tr><td>6</td><td>7</td><td>8</td></tr></table> | 0           | 1 | 2 | 3      | 4 | 5 | 6 | 7 | 8 | 3D Array<br>(3 <sup>rd</sup> order Tensor) |
| 0          | 1                                                                                                                                               | 2           |   |   |        |   |   |   |   |   |                                            |
| 3          | 4                                                                                                                                               | 5           |   |   |        |   |   |   |   |   |                                            |
| 6          | 7                                                                                                                                               | 8           |   |   |        |   |   |   |   |   |                                            |

# Adding Matrices and Vectors

- General rules
  - Matrices of same size can be added
  - A scalar can be added or multiplied with a matrix
- Extension = “Broadcasting”
  - Goal: add a vector to a matrix
  - ➔ Add ***b*** to every column of ***A***

$$C = A + B \text{ where } C_{i,j} = A_{i,j} + B_{i,j}$$

$$D = a \cdot B + c \text{ where } D_{i,j} = a \cdot B_{i,j} + c$$

$$C = A + \mathbf{b}, \text{ where } C_{i,j} = A_{i,j} + b_j$$

# Multiplying Matrices and Vectors

- For **vectors**: **inner product**, dot product, scalar product

$$x^\top y = \|x\|_2 \|y\|_2 \cos \theta \quad !$$

- Matrix multiplication** (from linear algebra)
  - Each value  $C_{i,j}$  is the **dot product** between row  $i$  of  $A$  and column  $j$  of  $B$

$$C = AB \quad C_{i,j} = \sum_k A_{i,k} B_{k,j} \quad !$$
$$= A_{i,:}^T B_{:,j}$$

- Element-wise multiplication** (Hadamard product)
  - ➔ matrices must have the same size

$$C = A \odot B \quad !$$

$$C_{i,j} = A_{i,j} B_{i,j}$$

# Multiplying Matrices and Vectors

- Matrix multiplication is **distributive**

$$A(B + C) = AB + AC$$

- Matrix multiplication is **associative**

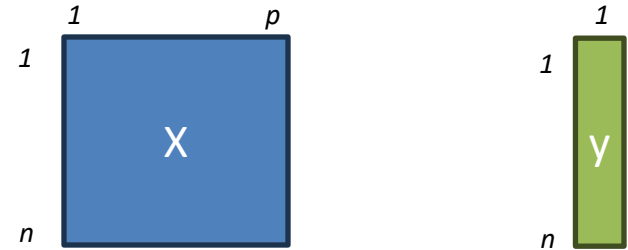
$$A(BC) = (AB)C$$

- Matrix multiplications are **not commutative**, which means  $AB$  is (in general) **not** equal to  $BA$ 
  - This is very easily forgotten 😊

$$AB \neq BA$$

# Application: Linear Regression

- Recap from Machine Learning:
  - Predict **output  $y$**  given some **inputs  $x_1, x_2, \dots, x_p$**
  - **Training dataset** of  $n$  measurements
  - ➔ Matrix  $X$  and vector  $y$
- Assumption: **linear model**
  - (ignoring the intercept)
- **Goal:** find vector  $\beta$



$$y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

$$y = X\beta$$



# Linear systems of equations

$$\boxed{Ax = b} !$$

- Linear system of equations
  - $\mathbf{x}$  is an unknown  $n$  dimensional vector

$A \in \mathbb{R}^{m \times n}$  is a known matrix

$\mathbf{b} \in \mathbb{R}^m$  is a known vector

- Alternative notation:
  - Each row of  $\mathbf{b}$  is a dot product

$$\left| \begin{array}{l} \mathbf{A}_{1,:} \mathbf{x} = b_1 \\ \mathbf{A}_{2,:} \mathbf{x} = b_2 \\ \dots \\ \mathbf{A}_{m,:} \mathbf{x} = b_m \end{array} \right|$$

- Alternative notation
- ➔ How to find the unknown  $\mathbf{x}$ ?

$$\left| \begin{array}{l} \mathbf{A}_{1,1}x_1 + \mathbf{A}_{1,2}x_2 + \dots + \mathbf{A}_{1,n}x_n = b_1 \\ \mathbf{A}_{2,1}x_1 + \mathbf{A}_{2,2}x_2 + \dots + \mathbf{A}_{2,n}x_n = b_2 \\ \dots \\ \mathbf{A}_{m,1}x_1 + \mathbf{A}_{m,2}x_2 + \dots + \mathbf{A}_{m,n}x_n = b_m \end{array} \right|$$

# Identity and Inverse Matrices

- For **scalars**, this would be trivial:

$$\begin{aligned}ax &= b \\ a^{-1}ax &= a^{-1}b \\ x &= a^{-1}b = \frac{b}{a}\end{aligned}$$

- What about **matrices**?

- $\rightarrow$  same method

- $A^{-1}$  is called the **inverse matrix**

$$\begin{aligned}Ax &= b \\ A^{-1}Ax &= A^{-1}b\end{aligned}$$

$$I_n x = A^{-1}b$$

$$x = A^{-1}b \quad !$$

- This requires the **identity matrix**  $I$

- This corresponds to the scalar 1

$$A^{-1}A = I_n \quad !$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

# Inverse Matrices

- **Key question:** does the inverse exist?

$$x = A^{-1}b$$

A **system of linear equations** can have either:

- *No solutions*
- *Exactly 1 solution*
- *Infinitely many solutions*

- If two solutions **x** and **y** exist, then any number of new solutions **z** can be created
- The **inverse exists** only if the set of linear equations has **exactly 1 solution**

$$z = \alpha x + (1 - \alpha)y$$

# Inverse Matrices

- For  $m$  equations (rows) and  $n$  unknowns (columns):
  - $n < m \rightarrow$  overdetermined, no solution
  - $n > m \rightarrow$  underdetermined, infinite number of solutions
  - $n = m \rightarrow$  **exactly one solution**
- **Unless multiple rows or columns are redundant  $\rightarrow$** 
  - This is called “**linear dependency**”

$$\left| \begin{array}{l} A_{1,1}x_1 + A_{1,2}x_2 + \cdots + A_{1,n}x_n = b_1 \\ A_{2,1}x_1 + A_{2,2}x_2 + \cdots + A_{2,n}x_n = b_2 \\ \cdots \\ A_{m,1}x_1 + A_{m,2}x_2 + \cdots + A_{m,n}x_n = b_m \end{array} \right|$$

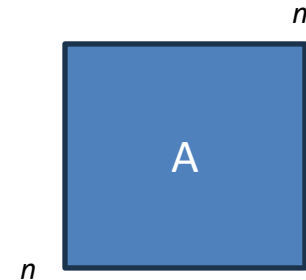
# Inverse Matrices

- Mathematically, the number of solutions can be determined with the **span** of the matrix
- A matrix multiplication can be rewritten as the **linear combination** on the right
  - Hence,  **$b$**  is a linear combination of the **columns** of  **$A$**
  - The **column span** of  **$A$**  is the set of all **points** that can be **reached by linear combinations** of the columns of  **$A$** .
- **→** if  **$b$**  is in the column span of  **$A$** , then  **$Ax=b$**  has a solution

$$Ax = \sum_i x_i A_{:,i}$$

# Inverse Matrices

- ➔ The **inverse  $A^{-1}$**  exists, if
  - $m=n$  (**square matrix**)
  - All columns are **linearly independent**
  - “Square non-singular matrix”



- A **square matrix** with **linearly dependent columns** is called a **singular matrix**

- ➔ not invertible!

- For **machine learning** problems, this is usually **not the case**

- Typically,  $n \gg m$
  - ➔ **other methods** required for finding the  $\beta$  vector

$$y = X\beta$$

$$\beta = X^{-1}y$$



# Norms

- **Goal** of norms: **measure length** of vector
- **General definition** of the  $L^p$  norm is given on the right
- Most widely used: **Euclidean norm** or  $L^2$  norm
  - The **squared  $L^2$  norm** can be calculated directly using the **inner product**
- Also frequently used:  $L^1$  norm

$$\|\mathbf{x}\|_p = \left( \sum_i |x_i|^p \right)^{\frac{1}{p}} \text{ for } p \in \mathbb{R}, p \geq 1 \quad !$$

$$\|\mathbf{x}\| = \|\mathbf{x}\|_2 = \sqrt{\sum_i (x_i)^2}$$

$$\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$$

$$\|\mathbf{x}\|_1 = \sum_i |x_i|$$

# Other Norms

- The  **$L^0$  “norm”** ( $p=0$ ) is the number of non-zero elements
  - Note: this is not a mathematically valid norm
- The  **$L^\infty$  norm** ( $p=\infty$ ) returns the **maximum** of the vector
- The **Frobenius norm** is an extension of the  $L^2$  norm to matrices

$$\|x\|_0 = \sum_i x_i^0$$

$$\|x\|_\infty = \max_i |x_i|$$

$$\|A\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

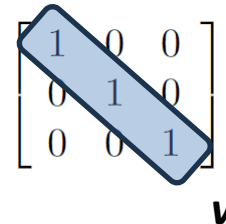
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# Agenda

- Fundamentals of vectors and matrices
- Special matrices
- Matrix decompositions
- Moore-Penrose-Pseudoinverse
- The Trace and the Determinant

# Diagonal matrices

- A **diagonal matrix** is all zero, except in the diagonal
  - Only the **vector of diagonal elements**  $\mathbf{v}$  contains information


$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{v}$$

- The notation on the right denotes a **diagonal matrix** with the diagonal elements in  $\mathbf{v}$
- **Multiplying a diagonal matrix with a vector** is computationally **efficient**
  - Only an element-wise multiplication is needed

$$\text{diag}(\mathbf{v})$$

$$\text{diag}(\mathbf{v}) \mathbf{x} = \mathbf{v} \odot \mathbf{x}$$

# Diagonal matrices

- **Inverting diagonal matrices** is also computationally **efficient**
  - Note: all  $v_i$  must be non-zero
- Applications:
  - **Speed up algorithms** by constraining matrices to be diagonal
  - **Gain theoretical understanding** by analyzing diagonal matrices

$$\text{diag}(\mathbf{v})^{-1} = \text{diag}([1/v_1, \dots, 1/v_n]^\top)$$

!

# Symmetric matrices

- A **symmetric matrix** is a matrix that is equal to its transpose
  - Surprisingly common case

$$A = A^T !$$

- **Example:** matrix of distances
  - $D_{ij} = D_{ji}$
  - → resulting matrix is symmetric

$$D_{ij} = \|x_i - x_j\|$$

# Orthogonal matrices

- **Two vectors are orthonormal** if
  - “ortho” → orthogonal (90° angle)
  - “normal” → length 1
  - i.e. if their inner product is 0
- An **orthogonal matrix** is a square matrix whose **rows are mutually orthonormal** and whose **columns are mutually orthonormal**
  - → trivial to invert
  - Simplifies processing

$$x^\top y = \|x\|_2 \|y\|_2 \cos \theta$$

$$x^\top y = 0$$

$$\|x\|_2 = 1$$

$$A^\top A = A A^\top = I \quad !$$

$$A^{-1} = A^\top \quad !$$

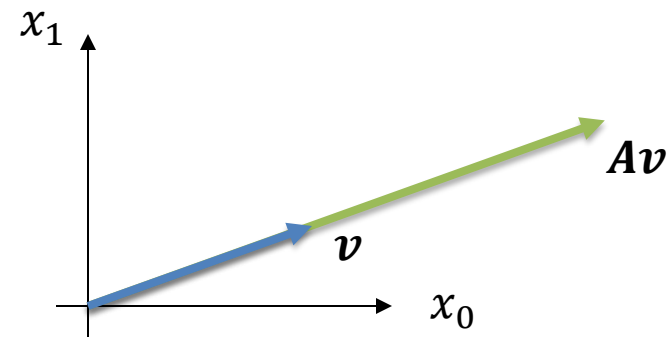
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# Eigendecomposition

- **Goal of eigendecomposition:** find **directions** that **remain unchanged** when multiplying with matrix  $A$ 
  - Mathematically: multiplying by  $A$  is equivalent to multiplying by a scalar  $\lambda$
  - $v$  = eigenvector (direction)
  - $\lambda$  = eigenvalue (factor)
- ➔ interpret Eigenvalues instead of full matrix  $A$
- In this course: focus on **real symmetric matrices**
  - In that case, the eigenvalues are guaranteed to exist and be real
  - Frequent case – at least in this course



$$Av = \lambda v$$

!

# Eigendecomposition

- **Eigenvectors** denote a **direction** in space
  - Eigenvectors are still eigenvectors, no matter how they are scaled
  - **Convention:** scale eigenvectors to **unit length!**
- A real symmetric matrix with  $n$  rows and columns has  $n$  real eigenvalues and  $n$  orthogonal eigenvectors

$$Av = \lambda v$$

$$\|v\| = 1 \quad !$$

$$\{v^{(1)}, \dots, v^{(n)}\}$$

$$\{\lambda_1, \dots, \lambda_n\}$$

# Eigendecomposition

- The **eigenvectors** can be concatenated into an **orthogonal** matrix  $\mathbf{V}$
- The **eigenvalues** can be concatenate into a vector  $\boldsymbol{\lambda}$
- The **matrix A** can then be rewritten in terms of  $\mathbf{V}$  and  $\boldsymbol{\lambda}$
- Hence, **multiplying by A** is equivalent to
  - **Rotate** by the orthogonal matrix  $\mathbf{V}$
  - **Multiply** (scale) by the eigenvalues
  - **Rotate back** into original coordinate system

$$\mathbf{V} = [\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}]$$

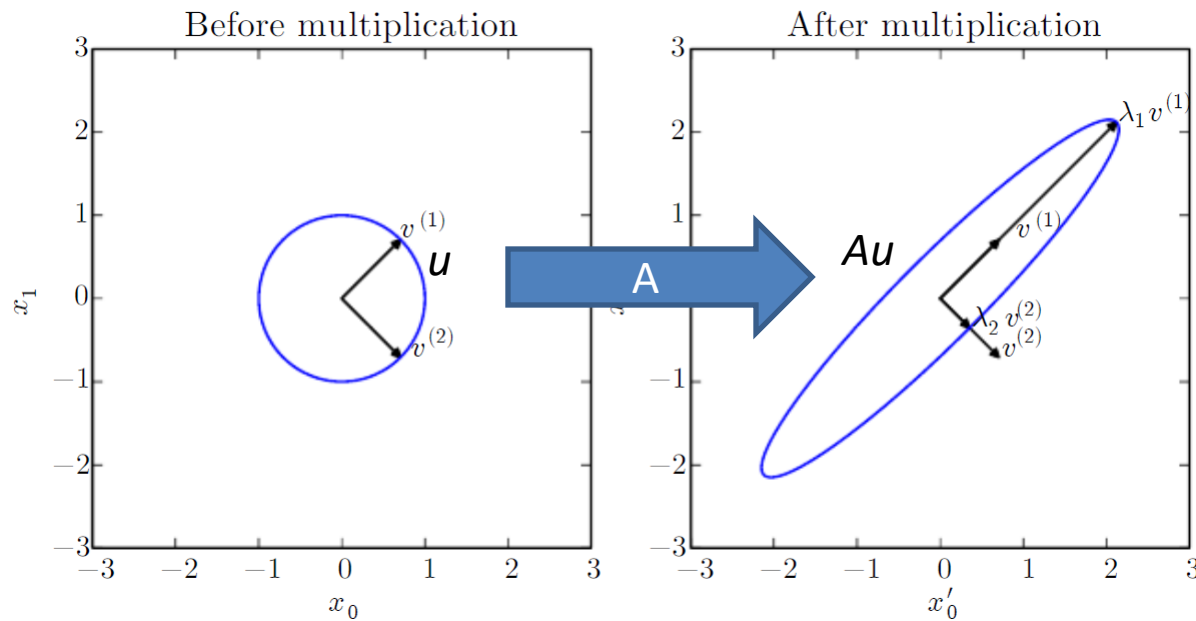
$$\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_n]^\top$$

$$\mathbf{A} = \mathbf{V} \text{diag}(\boldsymbol{\lambda}) \mathbf{V}^{-1} \quad !$$

# Eigendecomposition

- This can be visualized as shown below

$$A = V \text{diag}(\lambda) V^{-1}$$



*DL, Fig. 2.3*

# Eigendecomposition

- **Application: Interpretation** of the effect of a matrix  $A$
- **Application: the optimization problem** on the right can be solved with eigendecomposition
  - If  $x$  is a unit eigenvector of  $A$ , then
  - **Maximum**  $\rightarrow$  eigenvector belonging to the **maximum eigenvalue**.
  - **Minimum**  $\rightarrow$  eigenvector belonging to the **minimum eigenvalue**.

$$f(x) = x^T A x \quad \text{subject to } \|x\|_2 = 1$$

!

$$\begin{aligned} Ax &= \lambda x \\ x^T Ax &= x^T \lambda x \\ &= \lambda x^T x = \lambda \end{aligned}$$

# Singular Value Decomposition (SVD)

- **SVD → Generalization** of the eigendecomposition
  - **Every real matrix** has an SVD!
- **$U$** : orthonormal matrix
- **$D$** : diagonal matrix of “singular values”
- **$V$** : orthonormal matrix
- Again: rotation → multiplication by diagonal matrix → rotation
- **Application**: like eigendecomposition, but **more general**

$$A = V \operatorname{diag}(\lambda) V^{-1}$$



$$A = U D V^{\top}$$

# Agenda

- Fundamentals of vectors and matrices
- Special matrices
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- The Trace and the Determinant

# The Moore-Penrose Pseudoinverse

- **Problem of the inverse** → **does not exist** for many (important) cases
  - The **inverse  $A^{-1}$  exists**, if
  - $m=n$  (**square matrix**)
  - All columns are **linearly independent**
- **Options**
  - **$A$  has more (independent) rows than columns** → more equations than unknowns → no solution
  - **$A$  has less (independent) rows than columns** → more unknowns than equations → many solutions
- **Example: Linear regression**
  - 5 measured variables = columns
  - 100 observations = equations

$$Ax = y$$



# The Moore-Penrose Pseudoinverse

- **Approach:** find some matrix **B** that results in a reasonable solution
  - Ideally close to  $A^{-1}$
- **Moore-Penrose Pseudoinverse**
  - Definition:
  - “Pseudoinverse” = not true inverse, but closely related
- **Calculation:** with help of **SVD**
- **Inverting  $D$  is trivial** because it is diagonal

$$d_{ij}^+ = 1/d_{ij}$$

$$Ax = y$$

$$x = By$$

$$A^+ = \lim_{\alpha \searrow 0} (A^T A + \alpha I)^{-1} A^T \quad !$$

$$A = UDV^T$$



$$A^+ = VD^+U^T \quad !$$

# The Moore-Penrose Pseudoinverse

- The (true) **inverse**  $A^{-1}$  is defined as shown on the right

$$A^{-1}A = I$$

- For the **pseudo-inverse**:

$$A^+ = VD^+U^\top \quad A = UDV^\top$$

$$\begin{aligned} A^+A &= VD^+ \boxed{U^\top U} DV^\top \\ &= V \boxed{D^+ D} V^\top \\ &= \boxed{VV^\top} \\ &= I \end{aligned}$$

# The Moore-Penrose Pseudoinverse

- What does the pseudo-inverse result in?
- If **more rows** than columns
  - Over-determined, no solution
  - ➔ returns solution with smallest possible squared error
- If **fewer rows** than columns
  - Under-determined , many solutions
  - ➔ returns the solution with the smallest possible Euclidean norm

$$\|Ax - y\|_2$$

$$x = A^+ y \text{ with minimal Euclidean norm } \|x\|_2$$

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- Fundamentals of vectors and matrices
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# The Trace Operator

- **Trace operator**  $\rightarrow$  **sum of all the diagonal elements**
  - Application: replace sums (e.g. over some error) with the trace
  - $\rightarrow$  simplifies equations (especially if many sums involved)
- **Example: Frobenius norm**
  - Original notation: double sum
  - With trace: no more sums

$$\text{Tr}(\mathbf{A}) = \sum_i A_{i,i} \quad !$$

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

$$\|\mathbf{A}\|_F = \sqrt{\text{Tr}(\mathbf{A}\mathbf{A}^\top)} \quad !$$

# The Trace Operator

- **Properties of the trace**

- Invariant to transpose
- Invariant to circular shifts

$$\text{Tr}(\mathbf{A}^T) = \text{Tr}(\mathbf{A})$$

$$\text{Tr}(\mathbf{AB}) = \text{Tr}(\mathbf{BA})$$

$$\text{Tr}(\mathbf{ABC}) = \text{Tr}(\mathbf{CAB}) = \text{Tr}(\mathbf{BCA})$$

$$\text{Tr}\left(\prod_{i=1}^n \mathbf{F}^{(i)}\right) = \text{Tr}\left(\mathbf{F}^{(n)} \prod_{i=1}^{n-1} \mathbf{F}^{(i)}\right)$$

- Trace of a scalar: the scalar itself
- **Trace = sum of all Eigenvalues**

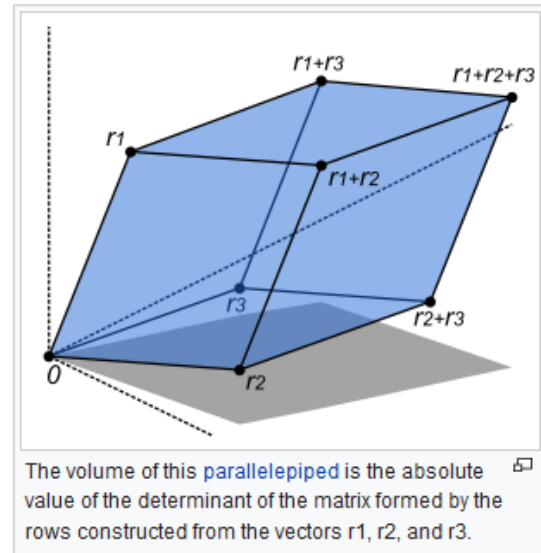
$$a = \text{Tr}(a)$$

$$\text{Tr}(\mathbf{A}) = \sum_i \lambda_i$$

# The Determinant

- Determinant: **product of all the eigenvalues**
  - Scalar value
  - If one eigenvalue = 0  $\rightarrow \det(A)=0$
  - Matrix is **invertible** if  $\det(A) \neq 0$
- Interpretation of determinant
  - Measure, **how much space expands** by multiplying with that matrix
  - If  $\det(A)=0$ , one dimension “collapses” and volume becomes 0
  - If  $\det(A)=1$ , volume is preserved

$$\det(A) = \prod_i \lambda_i !$$



# Agenda

- Fundamentals of vectors and matrices
- Special matrices
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- Example: Principal Components Analysis (PCA)



# Summary: Linear Algebra

- Systems of linear equations  $\mathbf{Ax} = \mathbf{b} \rightarrow$  solve with the inverse  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$
- Norms
  - $L^p$  norm 
$$\|\mathbf{x}\|_p = \left( \sum_i |x_i|^p \right)^{\frac{1}{p}}$$
  - Frobenius Norm 
$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{Tr}(\mathbf{A}\mathbf{A}^\top)}$$
 where trace  $\text{Tr}(\mathbf{A}) = \sum_i A_{i,i}$
- Special matrices
  - Orthogonal matrices  $\rightarrow \mathbf{A}^{-1} = \mathbf{A}^\top$
- Eigendecomposition
  - Eigenvalues  $\lambda$  and eigenvectors  $\mathbf{v}$  are solutions to  $\mathbf{Av} = \lambda\mathbf{v}$
  - The Eigendecomposition of  $\mathbf{A}$  is  $\mathbf{A} = \mathbf{V}\text{diag}(\boldsymbol{\lambda})\mathbf{V}^{-1}$  where  $\mathbf{V} = [\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}]$   
 $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_n]^\top$
- Singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ 
  - Generalization of Eigendecomposition to non-square matrices
- Moore-Penrose pseudoinverse
  - Generalization of inverse 
$$\mathbf{A}^+ = \lim_{\alpha \searrow 0} (\mathbf{A}^\top \mathbf{A} + \alpha \mathbf{I})^{-1} \mathbf{A}^\top$$
  - Typically calculated with SVD 
$$\mathbf{A}^+ = \mathbf{V}\mathbf{D}^+ \mathbf{U}^\top$$

Note: this is not a complete summary, only a very brief overview!

# Semester Plan

